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Special Course on Missile Aerodynamics

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NORTH ATLANTIC TREATY ORGANIZATION

ADVISORY GROUP FOR AEROSPACE RESEARCH AND DEVELOPMENT

(ORGANISATION DU TRAITE DE L'ATLANTIQUE NORD)

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SPECIAL COURSE

ON

MISSILE AERODYNAMICS

The material assembled in this book was prepared under the combined sponsorship of the Fluid Dynamics Panel, the von Kármán Institute and the Consultant and Exchange Program of AGARD and was presented as an AGARD Special Course at the von Kármán Institute, Rhode-Saint-Genèse, Belgium on 30 March—3 April 1987, and as Short Courses at Athens, Greece on 18—19 May 1987 and at Ankara, Turkey on 21—22 May 1987.

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PREFACE

This volume is a compilation of the edited proceedings of the "Missile Aerodynamics" course, held at the Von Kármán Institute (VKI) in Rhode-Saint-Genèse, Belgium, from March 30 to April 3, 1987. A condensed version of this course has been presented at the Air Academy in Athens, Greece, 18—19 May 1987 and at the Development and Research Department in Ankara, Turkey, 21—22 May, 1987.

This series of lectures supported by the AGARD Fluid Dynamics Panel and the Von Kármán Institute followed previous courses organized at VKI: 1974 (VKI LS67), 1976 (VKI LS88) and 1979 (AGARD LS98). This course was intended for practical engineers and researchers, beginners or experienced professionals, for military and engineering teaching institutions. The speakers, from industry and research establishments, paid particular attention to illustrate their presentation with numerous practical applications.

In recent years remarkable progress has been made in the field of tactical missile aerodynamics by theoretical and by experimental means and the objective of this course was to present the current state of the art in fundamental knowledge and in practical predictive methods (semi-empirical and numerical), with future trends. Special attention was focussed on nonlinear aerodynamics and unconventional configurations such as airbreathing missiles. For the first time numerical methods based on the resolution of the Euler equations with flow separation vortices were included in a course specific to missiles. In addition to the general aspects of missile aerodynamics, the course also dealt with particular problems such as the aerodynamics of air intakes, kinetic heating and base flows. The design of the next generation supersonic and hypersonic missiles was discussed in the last lecture.

We want to thank all the speakers for their outstanding work and AGARD and VKI for the organization of this course. Our thanks also go to the local coordinators in Athens and Ankara for their hospitality.

***

Ce volume regroupe les notes concernant le cours "Aérodynamique des Missiles" présenté à l'Institut Von Kármán (VKI) de Rhode-Saint-Genèse, Belgique, du 30 Mars au 3 Avril 1987 et dont une version condensée a été présentée à l'Académie de l'Air d'Athènes, Grèce, les 18—19 Mai 1987 et au Département Recherche et Développement d'Ankara, Turquie, les 21—22 Mai 1987.

Ce cycle de conférences conçu et réalisé sous l'égide du Panel de Dynamique des Fluides de l'AGARD et du VKI faisait suite à des cours similaires organisés au VKI en 1974 (VKI LS67), 1976 (VKI LS88) et 1979 (AGARD LS98). Ce cours était destiné aux ingénieurs de l'industrie et aux chercheurs, débutants ou confirmés, ainsi qu'aux militaires et écoles d'ingénieurs. Les conférenciers appartenant au secteur industriel et à des organismes de recherche s'attachèrent à illustrer leur présentation avec de nombreuses applications pratiques.

Des progrès considérables ayant été réalisés ces dernières années dans le domaine de l'aérodynamique des missiles tactiques aussi bien par voie théorique que par voie expérimentale, ce cours avait pour but de présenter l'état actuel des connaissances fondamentales et des méthodes de calcul utilisées (semi-empiriques et numériques) avec leur évolution future. Une attention toute particulière a été apportée aux effets non-linéaires et aux configurations non conventionnelles telles que les missiles aérobies. Pour la première fois les méthodes numériques basées sur la résolution des équations d'Euler avec prise en compte de structures tourbillonnaires ont été développées dans un cours spécifique aux missiles. Outre les aspects généraux de l'aérodynamique des missiles, les sujets traités au cours des conférences englobaient certains problèmes particuliers tels que l'aérodynamique des prises d'air, l'aérothermique et les écoulements de culot. La conception des missiles supersoniques et hypersoniques futurs a été présentée en conclusion de ce cours.

Nous tenons à remercier tous les conférenciers pour l'excellent travail qu'ils ont accompli ainsi que les organisateurs de l'AGARD et du VKI sans qui ce cours n'aurait pu avoir lieu, sans oublier les coordinateurs locaux de la Grèce et de la Turquie pour leur hospitalité.

R.G.Lacau
Directeur du cycle de conférences

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CONTENTS

AERODYNAMIC HEATING OF MISSILES

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SUMMARY

This paper presents the results of investigations of the aerodynamic heating of missiles. First the applied basic semiempirical methods to determine the flow field about the isolated components of missiles are described and some possibilities to calculate the heat flux in the boundary layer are specified for different pointed or blunted body noses, such as conical, ogival and hemispherical noses. The metbods are valid for laminar and turbulent flow properties and in the transition region between laminar and turbulent. The calculated heattransfer factors or Stanton numbers are compared with experimental results. The agreement between theoretical and measured values is good.

In the second part of the paper a theoretical method for the calculation of the flow field around missile configurations is presented. This method is based on the parabolization of the compressible, stationary Navier-Stokes equations (PNS-eq.) and limited to supersonic Mach numbers. The quality of the results is demonstrated by comparison to corresponding test results of other authors.

Finally a computational method is described to calculate the local time-dependent temperatures at the surface or inside the body. The configuration is divided into a number of volume elements, thus permitting numerical solution of the time varying equation of heat transfer. The resulting temperatures for some special configurations are plotted and when possible compared with experimental results.

1. INTRODUCTION

The knowledge of aerodynamic heating of missile-surface and of the corresponding increase of missile temperature may be very important during the design of missiles. The heating of the missile-body is very often a determining factor in the selection of materials, in order to ensure that the structure and the internal equipment of the missile, such as seeker, guidance control, power supply, warhead or propellant will not become too warm during its mission. Aerodynamic heating even may become critical for the operational efficiency of a missile, especially for an air-to-air-missile with an infrared-seeker and Mach numbers up to 4 and sometimes even more.

Figure 1 shows the theoretical stagnation temperature as a function of Mach number and altitude, indicating a large increase of stagnation temperature T_0 with increasing Mach number M and a slight decrease of T_0 with increasing altitude H. These stagnation temperatures represent a rough measure of the maximum temperatures induced by aerodynamic heating, the real maximum temperatures however are fortunately below the stagnation temperatures. By real gas effects the exact stagnation temperatures are lower than the values of fig. 1, which is valid for ideal gas.

The prediction of aerodynamic heating is possible by wind tunnel tests or by computational methods which include a wide spectrum of methods from approximate empirical methods to exact solutions of the full Navier-Stokes equations. A survey and introduction on common methods used to predict aerodynamic heating of missiles is given by Neumann and Hayes in [0], describing semiempirical methods, theoretical methods and test methods.

2. CALCULATION OF AEROKINETIC HEATING AND FLOW PROPERTIES

2.1 Semiempirical Method

The method to determine the heat flux between the boundary layer and the missile-surface is based on the assumption that the calculation may be applied to isolated components of the missile. The body is considered as a cylinder with either a hemispherical, conical or ogival nose, where the cone or the ogive may be pointed or blunted. The wing and tail are considered as flat plates with wedged or hemicylindrical noses. A summary of the possible configurations and the different flow states is shown in figure 3.

The considered velocities may be from subsonic up to hypersonic, but the results will be the better, the higher the Mach number is. The applied methods are valid for the range from zero up to about twenty degrees incidence with the restriction that no vortex or shock induced change of the heat transfer is considered.

It is important to note that at the junction between body and wing or tail, the induced vortices and shocks develop very high rate of kinetic heating. However, this effect is not considered in this method.

The heat flux $\dot{q}$ between the surrounding flow and the surface of a missile is induced by the gradient of the fluid temperature at the solid wall $(\partial T/\partial y)_{y=0}$ according to Fourier's law of heat conduction $\dot{q} = \lambda\ (\partial T/\partial y)_{y=0}$. Figure 2 shows characteristic temperature profiles in the boundary layer with cooled, adiabatic and heated wall. Instead of using $\partial T/\partial y$, the heat flux can be expressed by the difference $T_r - T_w$ between the adiabatic wall temperature T_r (= recovery temperature) and the actual wall temperature T_w as $\dot{q} = \alpha_w(T_r - T_w)$. The semiempirical method presented in this paper is based on this equation. To calculate the heat flux it is necessary to determine the quantities listed in figure 4. In order to avoid the complicated boundary layer calculations, empirical approximations are used for the estimation of the flow parameters T_r, α_w, T_δ, M_δ.

In figure 4 the subscript δ indicates a boundary layer condition, whereas the subscript w denotes a wall condition. The most important quantities to determine the heat flux are the heat-transfer factor α_w or the corresponding, nondimensional, so called Stanton number St and the recovery factor r, which is the ratio of temperature increase due to friction to the increase due to compression. To calculate the Stanton number we must determine the flow field and especially the pressure distribution in the nearfield of the body. In figure 5 some methods to calculate the local pressure distribution for several configurations can bee seen. For the most interesting case of supersonic and hypersonic Mach numbers we calculate the pressure distribution by the following methods:

For blunt bodies and pointed bodies with detached nose shocks we determine the pressure of the body surface by the so-called "modified Newtonian theory", which supplies very good results over the forward portion, but predicts free stream pressure at the shoulder.

Therefore we use the so-called "blast wave analogy" near the shoulder. This method is not applicable either in the nose region, where the details of the flow are important or far downstream where the shock wave decays to a Mach wave. The limiting value between the modified Newtonian theory and the blast wave theory is that point on the surface from which on the Newtonian pressure is less than the blast wave value. Farther downstream we use the blast wave theory as long as the pressure is higher than the ambient pressure.

For pointed bodies with attached nose shock we use equations, tables and charts for compressible flow (i. e. see [1]) in the nearfield of the stagnation point. Near the junction between nose and cylindrical body we determine the surface pressure from tables of supersonic flow over cone cylinder (i. e. see [2]).

A very important criterion for the heat flux is the physical effect whether the flow is laminar or turbulent. This difference determines the size of the Stanton number, which for turbulent flow may be up to ten times the value for laminar flow, whereas the recovery factors differ only slightly ($r \approx 0.82$ up to 0.85 for laminar flow and $r \approx 0.88$ for turbulent flow). The ratio between turbulent and laminar heat transfer factors increases with increasing Reynolds number. This means that for many configurations the heat transfer factor has its highest value far downstream from the stagnation point. This might be surprising in the first moment, but we should keep in mind this physical fact, when we later calculate the temperature distribution on the surface of the body.

The beginning of flow transition is influenced by several parameters. Some of the governing parameters are the flight Mach number, the local Mach number and the local Reynolds number at the outer edge of the boundary layer, but besides these there are additional parameters such as pressure gradient in the flow direction, geometrical shape of the nose, surface roughness or temperature ratio between surface and flow. In the literature we often find the local momentum thickness Θ and the displacement thickness δ^* instead of the local Mach- and Reynolds numbers.

In our digital program we use experimental results from H. Schlichting [3], K. F. Stetson [4], R. W. Detra [5], K. R. Czarnecki [6] and L. D. Wing [7] for the beginning of flow transition. Moreover we have to remember that the extent of the transition region between lamiar and full turbulent flow increases rapidly with increasing Mach number, where the transition region is usually considered to be characterized by the intermittent appearance of turbulent spots which grow as they move downstream until they finally merge into one another to form the turbulent boundary layer. This "spot theory" from H. W. Emmons [8] is extended to flows on blunt bodies by K. K. Chen and N. A. Thyson [9] and is fitted in the computer program.

To calculate the local Stanton number and recovery factor we use a procedure from L. D. Wing [10] for ogival noses and a method from E. R. Van Driest [11] for all other configurations. According to Van Driest's method the Stanton number depends on the geometrical shape and on the local Reynolds number, which is determined with the flow parameters at the outer edge of the boundary layer. The Stanton number is inversely proportional to the square root of the Reynolds number in laminar flow and also inversely proportional to the fifth root of the local Reynolds number in turbulent flow. It should be noted here that Van Driest's method neglects real gas effects.

A typical result for the Stanton number on a flat plate is shown in figure 7, where the local flow parameters in the boundary layer are varied. In figure 8 we see the local heat flux for laminar turbulent flow state and in the transition region of a sphere.

For a pointed tangent ogive nose with attached shock the flow parameters are calculated by subdividing the ogival nose into a short pointed conical region followed by several truncated cones. The boundary layer heat transfer rate and shear stresses at the wall are calculated by means of the Eckert and Tewfik [12] adaption of Lee's momentum integral equation and the use of Reynolds analogy for the laminar case, and the flat plate reference enthalpy method described in [13] (also applying Reynolds analogy) for the turbulent boundary layer case.

2.2 Theoretical Method (PNS-Method)

Theoretical calculations of flow fields around solid bodies at high supersonic or hypersonic speeds and the calculation of heat transfer at all Mach numbers must be based on the complete conservation equations, including viscous effects and careful treatment of the boundary layer. The consideration of complex vortex- and shock-systems is important, as they may effect extremely high local heat flux to the body surface. Figure 9 gives a classification of approximation levels of theoretical methods for the calculation of flow fields. Heat transfer calculations are possible only by the methods of level 1 to 3.

Full 3-D Navier-Stokes calculations of flow fields around realistic configurations are almost impossible with the available generation of computers. Therefore it is reasonable that efficient theoretical methods are based on approximations to the Navier-Stokes equations.

The "Thin-Layer"-approximation is able to give very good results [18], [19], but needs big computers [20] and has some problems with viscous effects at curved walls [21]. The computational expenditure is close to that of full Navier-Stokes solutions, using time-step-integration.

Using the parabolized Navier-Stokes equations (PNS) the amount of needed CPU-time and store capacity can be reduced to reasonable values. PNS-approximations are limited to supersonic flow calculations without recirculation, open flow separations however (vortex sheet separation) are included. Compared to methods solving the complete time-dependent conservation equations, the advantage of PNS-methods is given by the spatial integration in direction of the main flow.

Successfull applications of PNS-methods, based on the approximations of Vigneron et al. [22] and Schiff and Steger [23] include 3-D flow calculations for simple reentry bodies [24], for the US space shuttle [25], [26], [27], and for a supersonic fighter [28]. All cited applications used PNS-methods based on finite difference discretization. The presented Dornier method by Rieger [29] is based on a finite volume formulation.

Starting with the integral form of the conservation equations (s. figure 10), the finite volume formulation of the conservation equations is defined (see fig. 11, 12; notation after [30]). The system of equations has to be completed by the equations of a perfect gas, of Fourier's-law for molecular transfer of energy (heat conduction), and of Newtonian fluid (specification of Newtonian stress tensor). The influence of the temperature on the dynamic viscosity is used after Sutherland.

Details of the mathematical formulations and of the parabolization of the equation system are given by Rieger [29]. Figure 13 gives a characterization of the purpose, problem and measures of the parabolization. It has been proofed [22] that a good approximation of the pressure gradient in the subsonic layer is important and neccessary for accurate results, except for some hypersonic problems [31].

In the "sublayer"-method (see fig. 14), extended by Schiff and Steger [23] to non-iterative PNS-methods, the pressure gradient in integration direction of the subsonic layer is taken from the bordering supersonic flow. This assumption seems to cause a consistency problem and a limitation of the minimum integration intervals for numeric stability (see [23], [32], [33]).

Therefore the Dornier PNS-method is based on the Vigneron-approximation [22], using a portion of the pressure gradient which gives no mathematical problems and no limitation of the minimum integration interval. This is important for the accurate flow field calculation at pointed body noses.

In order to reduce the amount of CPU-time and of store capacity, the presented PNS-method assumes (see fig. 15) that the planes x^1 = const. of the curved coordinate system $x^i = (\xi, \eta, \zeta)$ with $(i = 1, 2, 3)$ are identical with the planes $x^{1'}$ = const. of a fixed cartesian coordinate system $x^{i'} = (x, y, z)$ with $(i' = 1', 2', 3')$.

2.3 Results of Aerokinetic Heating and Flow Field Calculations

Semiempirical method:

Figure 16 shows calculated heat transfer factors or Stanton numbers for a sphere (M_∞ = 8.9) compared with experimental results. The agreement between the two theoretical methods and between theory and experiment is good. In figure 17 we see the results for a blunted cone (M_∞ = 7.0). Here we have great difference between the same methods as in figure 16, especially in the region of the junction from spherical to conical body. Compared with measurements, we can see that the presented method is better than Lee's method. In figure 18 we see a blunted conical nose followed by a cylindrical body, the so-called AGARD-calibration-model HB-1. The agreement between calculalted and experimental values is very good. In figure 19 the heat transfer factor is shown for a pointed ogive nose followed by a cylindrical body. The angle of incidence is varied from -10 degrees up to

10 degrees. Near the stagnation point the heat transfer factor decreases downstream, but behind the station, where the transition begins, we recognize a very high increase in the heat transfer. The increase on the windward side is much higher than on the leeward side. The differences between windward and leeward sides are relatively weak on the clyindrical body. The agreement between theoretical and experimental results of the ogive nosed body is also good.

PNS-method:

For testing the PNS-method some typical configurations were used. Figure 20 shows calculated pressure coefficients compared to test results for the NASA-forebody No. 4 (after [34]) at $M = 1.7$, $\alpha = -5°$. The agreement between theoretical and test results is good. Comparisons of PNS-results and TNS-results of Haase [35] for a flat plate in supersonic flow ($M = 3.0$) proof good accuracy of the PNS-method (see fig. 21, 22). Figure 24 shows the comparison of the present PNS-method results with test result of Holden and Moselle [36], and with theoretical results of Hung and MacCormack [37] and Lawrence et al. [38] for a hypersonic compression ramp defined in figure 23.

An important testcase is given by the calculations for a pointed cone (at $M = 7.95$; $\alpha = 12°, 24°$) which has been tested by Tracy [39]. Figure 35 shows the shock position in the solution adaptive coordinate system after Haase [40] which agrees very well with the test results of Tracy. The pressure coefficients and the heat transfer coefficients (see fig. 26, 27) versus the circumferential angle are in very good agreement with the test results.

Figure 28 shows the geometry of an ogive for which some results of the time dependent temperature distribution are available [17]. Heat flux calculations by the Dornier-PNS-method and by the semiempirical method proof a good agreement of the calculated results (fig. 29) at $M = 1.99$ and $\alpha = 0°$, confirming the good experience with the semiempirical method.

3. CALCULATION OF LOCAL TEMPERATURES

3.1 Calculation method

If the Stanton numbers and the recovery factors are determined, the local time-dependent temperatures at the surface or inside the missile can be calculated. Therefore the configuration is divided into a number of volume elements, thus permitting numerical solution of the time varying equations of heat transfer. There is a physical requirement in the method, that the temperature is constant within a volume element. The temperature of an element depends on heat transfer, heat convection, picked up and reflected heat radiation, internal heat conduction and from internal losses. For any volume element the heat balance is expressed (see figure 30) and we obtain a system consisting of a number of first order differential equations.

For steady state flows, we can determine the temperatures by solving the system of equations iteratively. For intermittend states of flow, which are typical in missile flights, we solve the system of ordinary nonlinear differential equations by a fifth order Runge-Kutta-method. The computational organization of the digital program provides that the heat transfer factors α_w and the corresponding recovery temperatures T_{rec} for several Mach numbers and angles of attack are determined in a first step. The results will be used in form of input data in the second step, the computation of local temperatures. This simplification requires that the influence of temperature on the heat transfer factor is negligible.

It should be noted here that the time for computation of the local temperatures depends first of all on the number of volume elements. With 24 volume elements and a duration of flight of about 30 seconds in an example, we had about two minutes of computation time whereas the semiempirical calculation of aerodynamic heating runs very much quicker.

3.2 Results of Local Temperature Calculations

In figure 31 we see the time varying local temperatures on the surface of a spherically-nosed cylindrical body. The appropriate Mach-time-curve is typical for an air-to-air missile. The most important facts in this figure are that the station with the highest temperature of the body is not the stagnation point, but the station of the beginning of transition and that the maxim value comes chronologically clearly after the maximum of Mach number. Besides these effects we should notice that those stations with high heating during acceleration of the missile lose more temperature when the Mach number decreases. The heating of the cylindrical regions is weak compared with that of the stagnation region.

Figure 32 shows time-dependent temperatures at the stagnation point for one-layer-model and five-layer-model calculations. Here we can see that it is extraordinarily necessary to use a multi-layer-model. With an one-layer-model we determine temperatures which are too low during acceleration and too high during a following speed retardation. If we want to know the exact surface temperatures or internal temperatures the multi-layer-model is necessary except the heat conduction is relatively good (see figure 14). But we should remember that the materials of body noses are usually glass or something like that especially if we have infrared seekers.

The results in figures 31, 32 and 33 are valid for axisymmetric flow. In figure 34 the angle of incidence is 20 degrees. The temperature on the windward side is up to about 150 degrees higher than on the leeward side. This statement is only valid during acceleration of the missile. When the Mach number decreases,the temperature decreases too, where the reduction in temperature is higher for areas of higher temperature during accelerated flight.

In figure 35 some theoreticaly determined temperatures of a pointed tangent ogive nose are compared with experimental results, where the Mach number was constant for about half a minute. First of all we have to note that the differences between calculated and experimental results are not greater than about twenty degrees, which means a good agreement. It is nearly impossible to specify the reasons for the differences, but it seems important to show that the theoretical recovery temperatures after a relatively long time of mission are too low in the stagnation- and transition region, whereas more downstream (element C in figure 35) the agreement is very good.

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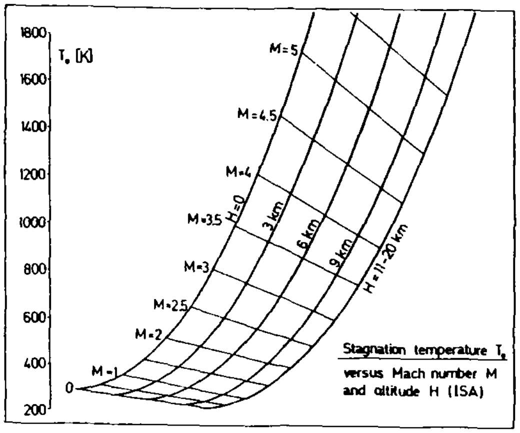


Figure 1

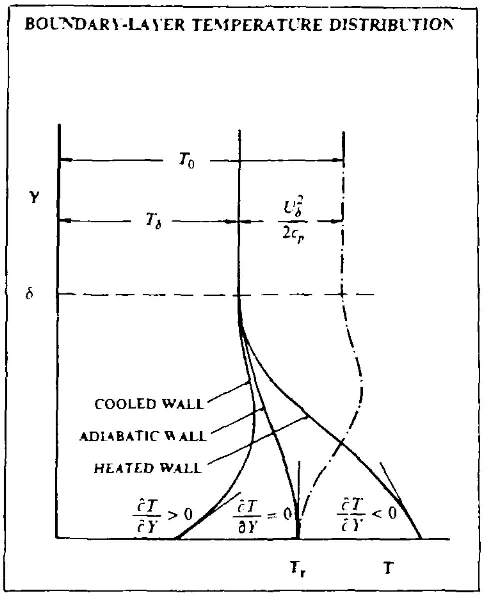


Figure 2

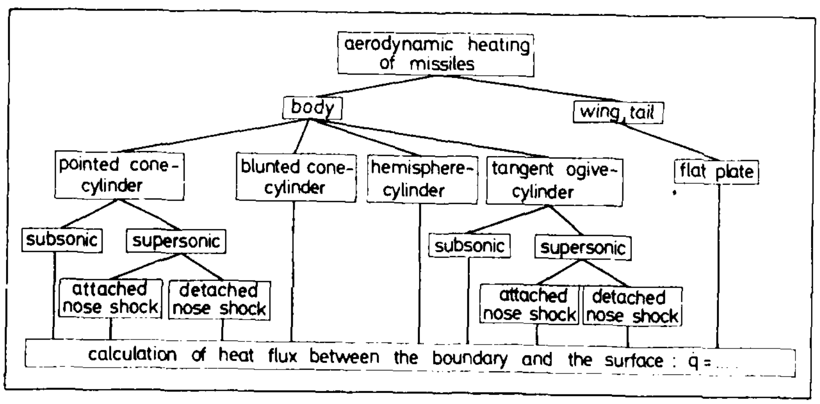


Figure 3

some general aerokinetic equations:

$$\boxed{\dot{q} = \alpha_W (T_r - T_W)}$$

α_W ..... heat-transfer coefficient $= f(H, M, \alpha, T_W, \text{shape})$

T_r .... recovery temperature in the boundary layer $= g(H, M, \alpha,$ flow state$)$

T_W .... local wall temperature

$$\boxed{T_r = T_\delta \cdot [1 + 0.5 \cdot r (\varkappa - 1) \cdot M_\delta^2]}$$

T_δ ...... temperature } at the outer edge of boundary layer

M_δ ..... Mach number } at the outer edge of boundary layer

r .... recovery factor

$\varkappa$ ...... adiabatic exponent

$$\boxed{St = \frac{\alpha_W}{\rho \cdot u \cdot c_p}}$$

St ..... Stanton number

ρ ..... density

u ..... velocity

c_p ..... specific heat coefficient

Figure 4

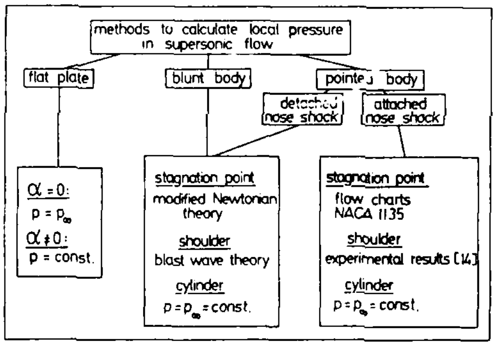


Figure 5

laminar Stanton number:

$$\boxed{St_{\delta,l} = f_l \cdot (Re_{x,\delta})^{-1/2}}$$

f_l, f_t .... coefficients depending on nose shape and Prandtl number

$$\boxed{St_{\infty,l} = f_l \cdot \left(\frac{u_\delta}{u_\infty}\right)^{1/2} \cdot \left(\frac{\mu_\infty}{\rho_\infty \cdot u_\infty \cdot x}\right)^{1/2} \cdot \left(\frac{\rho_\delta}{\rho_\infty}\right)^{1/2} \cdot \left(\frac{\mu_\delta}{\mu_\infty}\right)^{1/2} \cdot \left(\frac{c_{p,\delta}}{c_{p,\infty}}\right)}$$

turbulent Stanton number:

$$\boxed{St_{\delta,t} = f_t \cdot (Re_{x,\delta})^{-1/5}}$$

$$\boxed{St = f_t \cdot \left(\frac{u_\delta}{u_\infty}\right)^{4/5} \cdot \left(\frac{\mu_\infty}{\rho_\infty \cdot u_\infty \cdot x}\right)^{1/5} \cdot \left(\frac{\rho_\delta}{\rho_\infty}\right)^{4/5} \cdot \left(\frac{\mu_\delta}{\mu_\infty}\right)^{1/5} \cdot \left(\frac{c_{p,\delta}}{c_{p,\infty}}\right)}$$

Reynolds number:

$$\boxed{Re_{x,\delta} = \frac{\rho_\delta \cdot u_\delta \cdot x}{\mu_\delta}}$$

subscripts:

δ ..... boundary layer

∞ ..... infinity

l ...... laminar

t ...... turbulent

symbols:

c_p ... specific heat coefficient

u ... local velocity

x .... distance from stag.point

λ .... heat conduction coefficient

μ .... fluid viscosity coefficient

ρ .... density

Figure 6

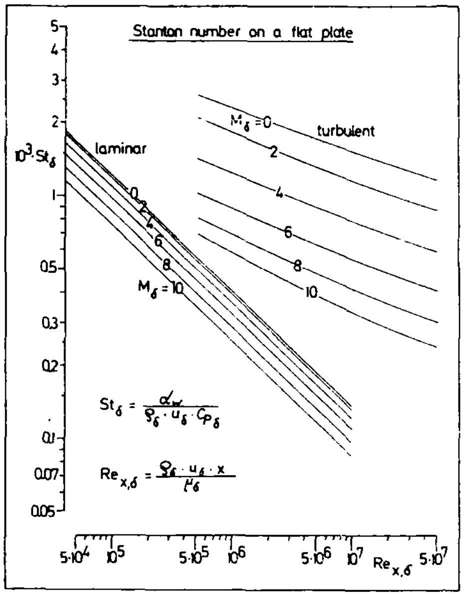


Figure 7

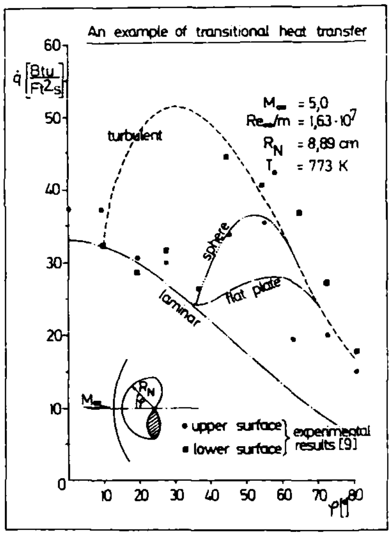


Figure 8

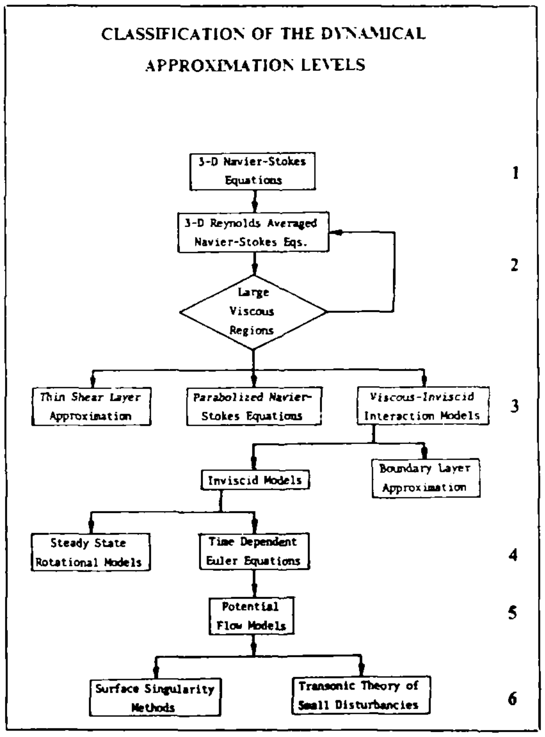


Figure 9

STEADY STATE FORMULATION
OF CONSERVATION EQUATIONS

$$\int_S (A\underline{v} + B) \cdot \underline{n}dS = 0$$

CONSERVATION OF MASS : $A \equiv \rho \qquad ; B \equiv 0$

CONSERVATION OF MOMENTUM : $A \equiv \rho\underline{v} \qquad ; B = -\underline{\underline{\sigma}}$

CONSERVATION OF ENERGY : $A \equiv E_{tot} = \rho e + \frac{1}{2}\rho\ (\underline{v}\cdot\underline{v})$

$B \equiv -\underline{\underline{\sigma}} \cdot \underline{v} + \underline{q}$

Figure 10

DIRECT FINITE VOLUME DISCRETIZATION

CONSERVATION OF MASS $\int_S \rho v^{l'} dS^k_{l'} = 0$

MEAN VALUE THEOREM OF INTEGRAL CALCULUS

$$\int_S \rho v^{l'} dS^k_{l'} = \sum_{k=1}^{3} \Delta_k\ [\overline{\rho v}^{l'}\ \delta S^k_{l'}] = 0$$

$$\Delta_k\ [\bar{f}^{i'} \delta S^k_{i'}] = [\ldots]_+ - [\ldots]_-$$

Figure 11

DIRECT FINITE VOLUME DISCRETIZATION

CONSERVATION OF MOMENTUM $\{\sum_{k=1}^{3} \Delta_k\ [(\rho v_{l'} v^{m'} + p\delta^{m'}_{l'} - \tilde{\sigma}^{m'}_{l'})\ \delta S^k_{m'}]\}\ \underline{g}^{l'} = 0$

CONSERVATION OF ENERGY $\sum_{k=1}^{3} \Delta_k\ [([E_{tot}+p]v^{m'} - \tilde{\sigma}^{m'}_{l'} v^{l'} + q^{m'})\ \delta S^k_{m'}] = 0$

ASSUMPTIONS:
- PERFECT GAS
- VALIDITY OF FOURIER LAW
- NEWTONIAN FLUID
- DYN. VISCOSITY DEPENDS ON TEMPERATURE AFTER SUHTHERLAND

Figure 12

PARABOLIZING FOR THE SUPERSONIC REGIME

PURPOSE : Suppression of elliptical influences, as stable integration is guaranteed for hyperbolic-parabolic systems of equations only

PROBLEM : Subsonic regime in boundary layer (upstream influence - elliptical influence)

MEASURES : Direction of integration approximately parallel to main direction of flow (e.g. x axis)

Neglecting of all viscous and diffuse termes in direction of integration

Figure 13

PRESSURE GRADIENT IN DIRECTION OF INTEGRATION

TOTAL NEGLECT (Lin & Rubin)

Limitation of applicability

"SUBLAYER"-APPROXIMATION (Lin & Rubin ; Schiff & Steger)

No normal pressure gradient in subsonic layer

Pressure gradient in direction of integration of subsonic layer taken from supersonic regime

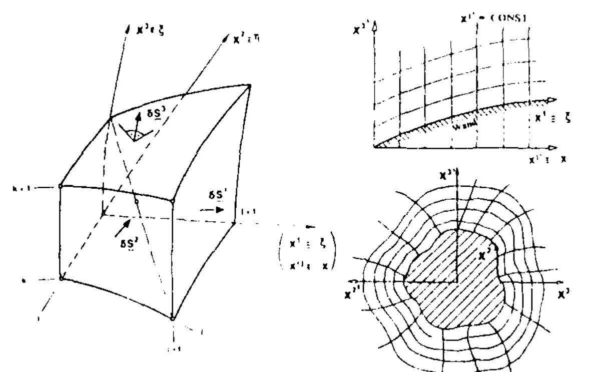
Consistency problem

Limitation of minimum interval of integration

"VIGNERON"-APPROXIMATION (Vigneron, Rakich & Tannehill)

Consideration of pressure gradient as far as mathematically possible

Figure 14



FINITE VOLUME AND COORDINATE SYSTEM

Figure 15

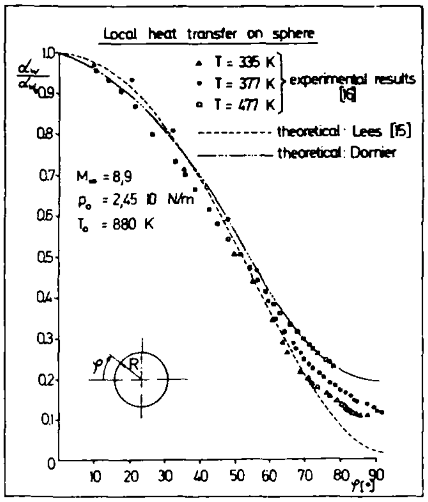


Figure 16

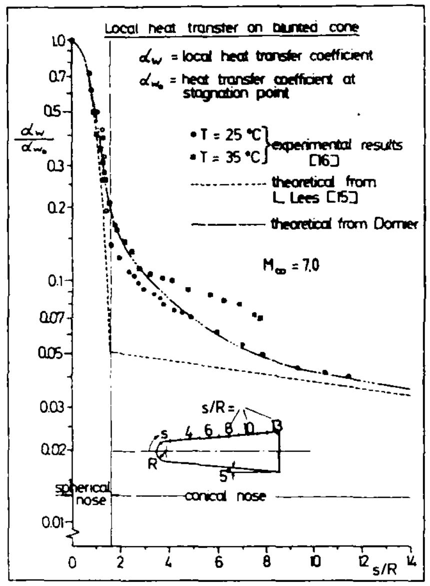


Figure 17

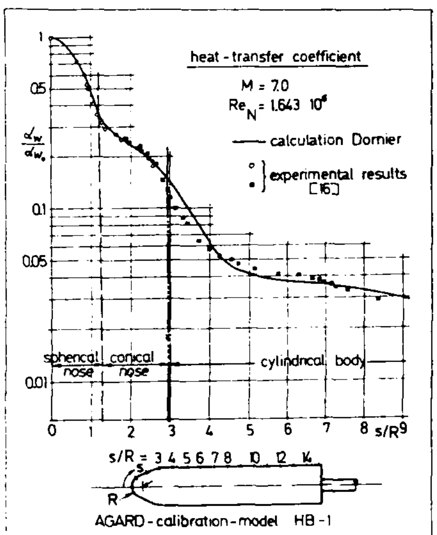


Figure 18

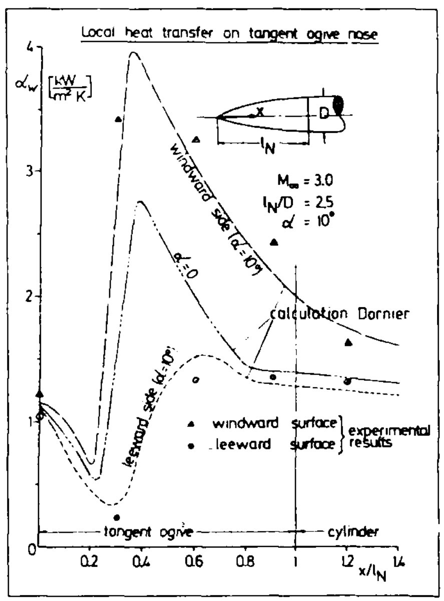
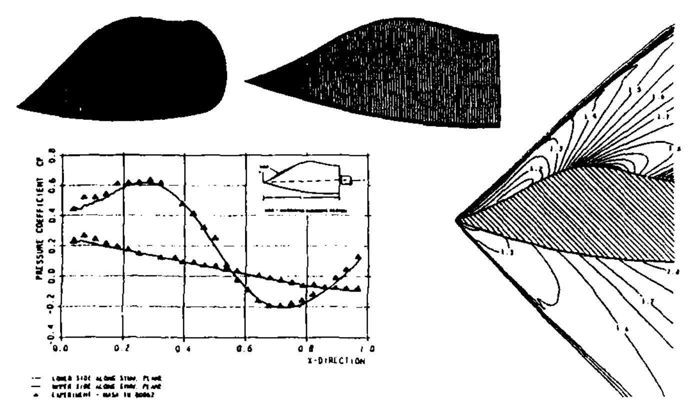


Figure 19

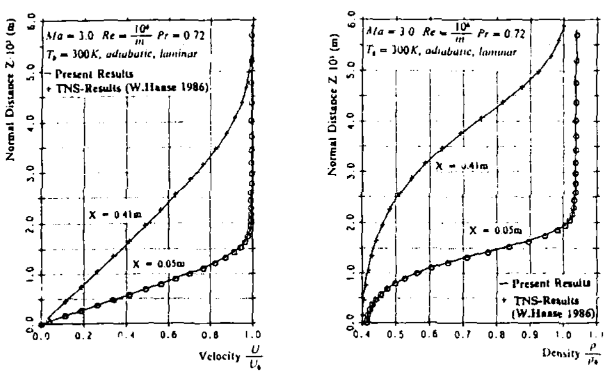


Wall Pressure Distribution in Symmetry Plane at Lower and Upper Side

Mach-Number Distribution in Symmetry Plane

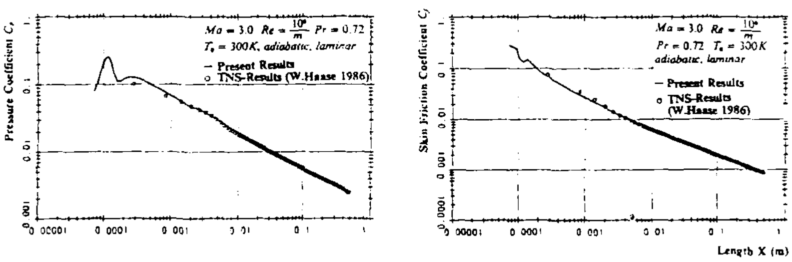
NASA - Forebody No.4 ($Ma = 1.7, \alpha = -5^o$)

Figure 20



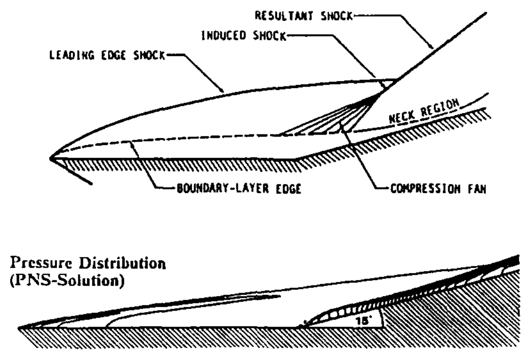
U-Velocity and Density Profiles for Supersonic Laminar Flow over a Flat Plate

Figure 21



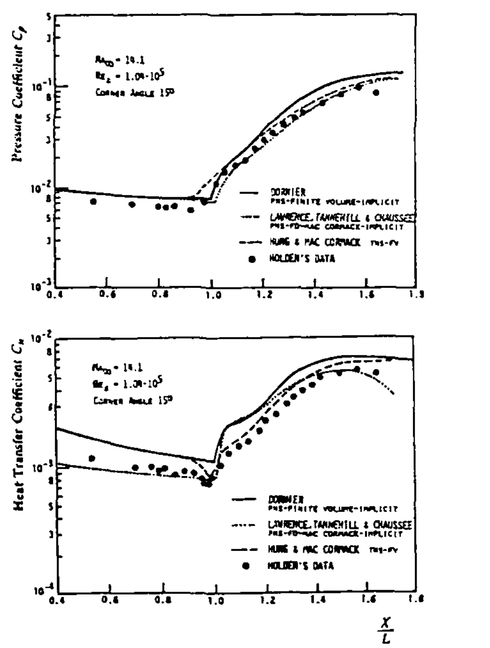
Wall Pressure and Skin Friction Coefficient for Supersonic Laminar Flow over a Flat Plate

Figure 22



Flow Topology for a Hypersonic Compression Ramp

Figure 23



Wall Pressure and Heat Transfer Coefficient Along Hypersonic Compression Ramp

Laminar Flow $Ma = 14.1$ $Re_L = 1.04 \cdot 10^5$ $Pr = 0.72$ $T_\infty = 72.2K$ $T_w = 297K$ $L = 0.439m$

Figure 24

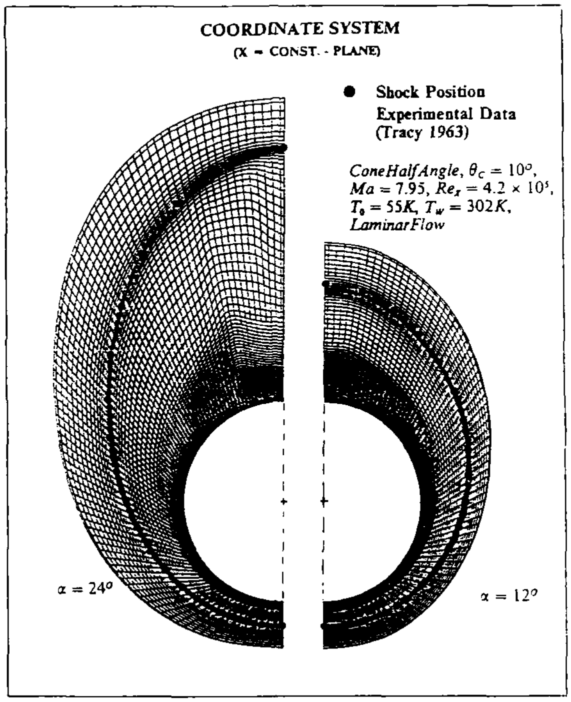
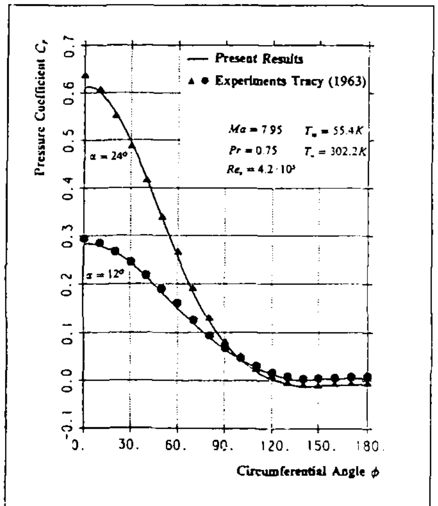
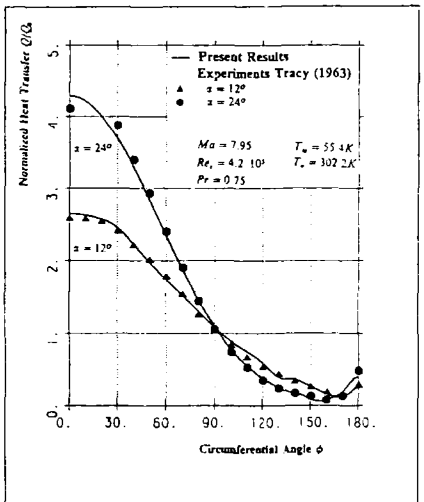


Figure 25



Wall Pressure Distribution Around a 10 Degree Half Angle Cone

Figure 26



Normalized Heat Transfer around a 10 Degree Half Angle Cone

(Q_0 - Heat Flux at 0 Degree Angle of Attack)

Figure 27

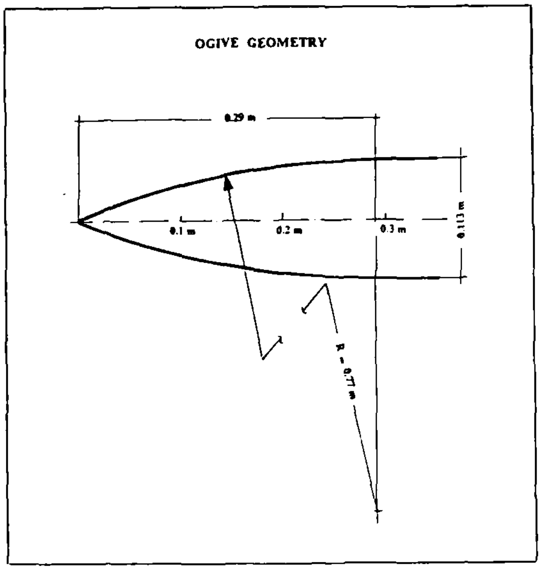


Figure 28

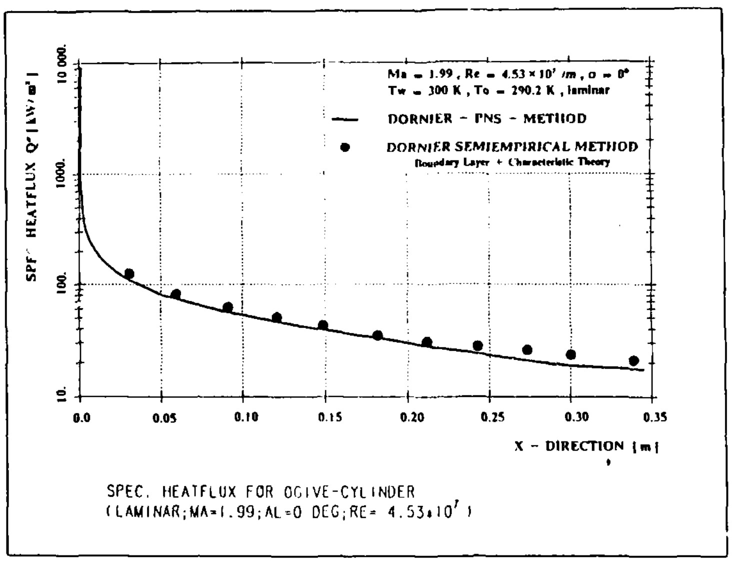


Figure 29

<u>heat balance for a volume element i :</u>

$$\boxed{m_i \cdot Cp_i \cdot \frac{dT_i}{dt} = \varepsilon_i \cdot F_i \cdot \sigma \cdot \left[\left(\sum_{j=1}^{n} B_{ij} \cdot T_j^4 \right) - T_i^4 \right] + \sum_{j=1}^{n} d_{ij} \, (T_j - T_i) + \dot{q}_{aero,i} + \dot{q}_V}$$

the physical meaning of these terms is:

$\varepsilon_i \cdot F_i \cdot \sigma \cdot \left(\sum_{j=1}^{n} B_{ij} \cdot T_j^4 \right)$ .... picked up heat radiation

$- \varepsilon_i \cdot F_i \cdot \sigma \cdot T_i^4$ .... reflected heat radiation

$\sum_{j=1}^{n} d_{ij} \cdot (T_i - T_j)$ .... heat conduction

$\dot{q}_{aero,i}$ .... heat transfer

$\dot{q}_V$ .... internal losses

Figure 30

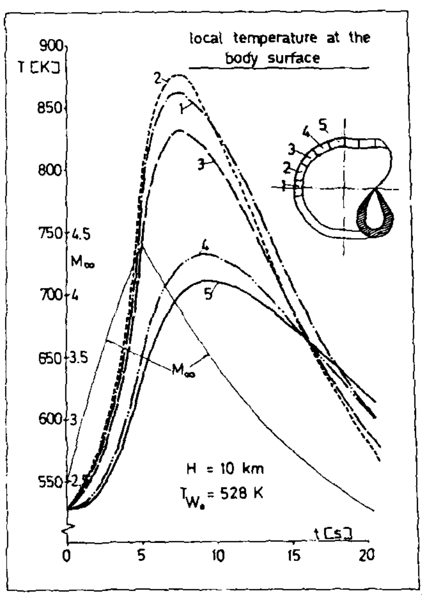


Figure 31

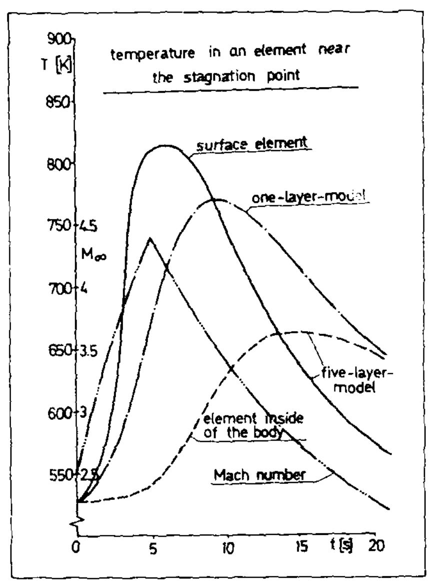


Figure 32

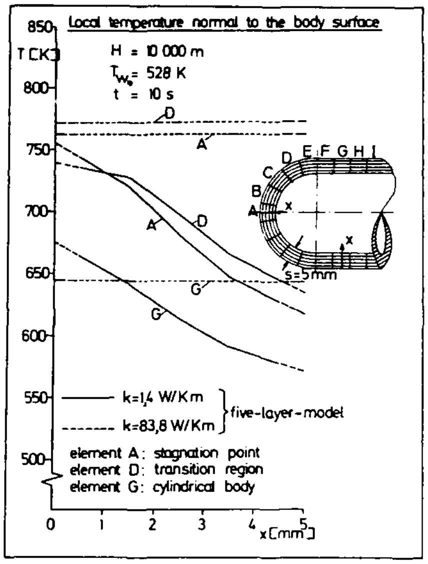


Figure 33

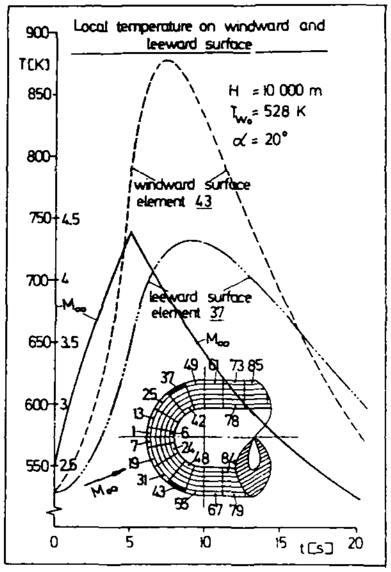


Figure 34

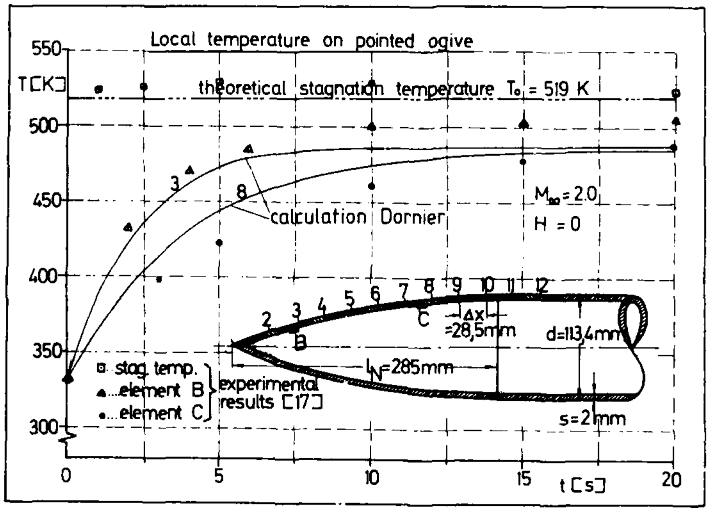


Figure 35